Cambridge (CIE) IGCSE Computer Science
Paper 1: Computer Systems (Set C)

Tuesday 6 May 2025

Morning (Time: 1 hour 45 minutes)

Total marks

/ 77

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].
- No marks will be awarded for using brand names of software packages or hardware.
- Calculators must not be used in this paper.

Scan here to mark your mock exam
or visit the mock exams landing page for this course

1 (a) A student is creating a range of documents for a school project.

The student records a podcast about computer science.

Describe how an **analogue sound wave** is **converted into digital form**.

....

....

....

(3 marks)

(b) **Tick (✓) one or more** boxes on **each row** to identify the effect(s) that each change will have on the sound file.

Change	File size increases	File size decreases	Accuracy increases	Accuracy decreases
Duration changes from 10 minutes to 20 minutes				
Sample rate changes from 44 kilohertz to 8 kilohertz				
Bit depth changes from 8 bits to 16 bits				

....

....

....

(3 marks)

2 (a) **State** what is given to a device when it connects to a wireless network using the SSID and correct password.

....

(1 mark)

(b) **Complete** the sentences about symmetric encryption.

Use the terms from the list.

Some of the terms in the list will not be used. You should only use a term **once**.

- **algorithm**
- **cipher**
- **copied**
- **delete**
- **key**
- **plain**
- **private**
- **public**
- **standard**
- **stolen**
- **understood**
- **unreadable**

The data before encryption is known as .... text. To scramble the data, an encryption ...., which is a type of ...., is used.

The data after encryption is known as .... text. Encryption prevents the data from being .... by a hacker.

(5 marks)

3 (a) **Identify two** reasons why errors can occur when using wireless technology to transmit data.

....

....

(2 marks)

(b) Four 7-bit binary values are transmitted from one computer to another. A parity bit is added to each binary value creating 8-bit binary values. All the binary values are transmitted and received correctly.

Identify whether each 8-bit binary value has been sent using **odd** or **even** parity by writing odd or even in the type of parity column.

8-bit binary value	Type of parity
01100100	
10010001	
00000011	
10110010	

....

....

....

....

(4 marks)

(c) The contents of three binary registers have been transmitted from one computer to another.

Even parity has been used as an error detection method.

The outcome after transmission is:

No errors have been detected in **Register A** and **Register C**.

An error has been detected in **Register B**.

Complete the parity bit for each register to show the given outcome

	Parity bit							
Register A		0	1	0	0	1	0	1
Register B		1	0	0	0	0	0	1
Register C		1	0	0	0	0	1	1

....

....

....

(3 marks)

4 (a) A manager at a company is concerned about a **brute-force attack** on its employee user accounts.

One possible aim for carrying out a brute-force attack is to **install malware** onto the company network.

Identify three types of malware that could be installed.

....

....

....

(3 marks)

(b) **Give two** security solutions that could be used to help **prevent** a brute-force attack being successful.

....

....

(2 marks)

(c) A student uses the internet for their schoolwork to research what is meant by **pharming**.

State the aim of pharming.

....

(1 mark)

5 (a) Layla uses her computer to create educational games

Layla makes use of **system software**.

One type of system software is the operating system.

Identify and **describe two functions** of an operating system.

....

....

....

....

....

....

(6 marks)

(b) Operating systems often provide **user interfaces**.

Describe the difference between a graphical user interface (GUI) and a command line interface (CLI) and give an example of where each might be used.

....

....

....

....

(4 marks)

6 (a) A user wants to connect their computer to a network.

Identify the component in the computer that is needed to access a network

....

(1 mark)

(b) **Identify** the type of address that is allocated to the component by the manufacturer, which is used to uniquely identify the device.

....

(1 mark)

(c) A dynamic internet protocol (IP) address is allocated to the computer when it is connected to the network.

Identify the device on the network that can connect multiple devices and automatically assign them an IP address.

....

(1 mark)

(d) A web server has an internet protocol (IP) address.

Give three characteristics of an **IP address**.

....

....

....

(3 marks)

(e) A user wants to connect their computer to a network.

A dynamic internet protocol (IP) address is allocated to the computer when it is connected to the network.

Describe what is meant by a **dynamic IP address**.

(3 marks)

7 A factory uses an automated system to control the temperature of its storage rooms.

The system consists of temperature sensors, a microprocessor, and actuators connected to heaters and cooling units. The sensors monitor the temperature and send the data to the microprocessor. The microprocessor processes the data and activates the heaters or cooling units based on pre-programmed rules.

Explain how the automated system maintains the desired temperature in the storage rooms.

(8 marks)

8 (a) A company uses robots in its factory to manufacture large pieces of furniture.

One characteristic of a robot is that it is programmable.

State two other characteristics of a robot.

....

....

(2 marks)

(b) **Give two** advantages to company employees of using robots to manufacture large pieces of furniture.

....

....

(2 marks)

9 (a) Binary is a number system used by computers.

Tick (✓) one box to show which statement about the binary number system is correct.

A. It is a base 1 system

B. It is a base 2 system

C. It is a base 10 system

D. It is a base 16 system

(1 mark)

(b) Denary numbers are converted to binary numbers to be processed by a computer.

Convert these three **denary numbers** to **8-bit binary numbers**.

50 ....

102 ....

221 ....

....

....

....

(3 marks)

(c) Two 8-bit binary numbers are given.

Add the two 8-bit binary numbers using binary addition.

Give your **answer in binary**. Show all your **working**.

$$\begin{array}{r} 0\,0\,1\,1\,0\,0\,1\,1 \\ +\,0\,1\,1\,0\,0\,0\,0\,1 \\ \hline \end{array}$$

....

....

(3 marks)

10 **State three different features** of a high-level programming language that a programmer could use to make sure that their program will be easier to understand by another programmer.

Give an example for each feature.

(6 marks)

11 Six statements are given about the operation of three different **types of storage**.

Tick (✓) to show which statements apply to each type of storage. Some statements **may apply to more than one** type of storage.

Statement	Magnetic (✓)	Optical (✓)	Solid state (✓)
this storage has no moving parts			
this storage uses a laser to read and write data			
this storage uses a read/write head			
this storage burns pits onto a reflective surface			
this storage uses NAND and NOR technology			
this storage stores data in tracks and sectors			

(6 marks)